

Technology Stack

Date	26 March 2026
Project Name	Credit Card Approval Prediction System

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Chosen to develop a responsive, user-friendly web interface for entering applicant details and viewing credit card approval predictions across different devices.
2	Backend / Server-Side	Python (Flask Framework)	Selected because Flask is lightweight, easy to develop with, and efficiently integrates machine learning models for real-time prediction.
3	Database / Data Storage	MySQL	Used to securely store applicant details, credit history, user information, and prediction records with reliable data management.
4	Cloud / Hosting / Deployment	Render (or Local Flask Server if not deployed)	Chosen to deploy the trained machine learning model for scalable, real-time credit card approval predictions accessible through a web application.

5	Version Control & CI/CD	Git & GitHub	Used for source code management, version control, collaboration, and project tracking throughout development.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, XGBoost	Used for data preprocessing, feature engineering, model training, evaluation, visualization, and credit card approval prediction.